MECH 632
Intermediate Spaceflight Dynamics

Review of Particle Mechanics

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Lesson Overview

1. Lesson Goals
2. Definitions of Key Terms
3. Review of Newtonian Mechanics
   - Kinematics of a Particle
   - Dynamics of a Particle

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Lesson Goals

- Review key terms and properties of matrices/vectors
- Review how to represent vectors in different coordinate systems
- Review hoe to differentiate vectors with respect to (w.r.t.) non-inertial reference frames
- Review Newtonian kinematics and dynamics

Scalar: Denoted by a variable a; a quantity characterized by magnitude only and is a single real number (e.g., mass, distance, energy, etc.).

Vector: Denoted by an overhead arrow $\vec{a}$; a quantity characterized by both direction and magnitude (e.g., velocity, force, momentum etc.).
Specifically *unit vectors*, with magnitude equal to one, are denoted by a hat $\hat{a}$.
Note: A vector is a physical entity and independent of any particular coordinate system used to represented such vector mathematically.

Matrix: Denoted by capital letter A; a rectangular array of elements which could be real numbers, complex numbers, or functions.

Matrix Properties

A matrix that has the same number of rows and columns is *square* (i.e. matrix of size $n \times n$).

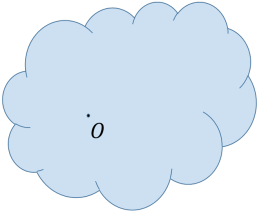
A matrix A is *symmetric* if $A = A^T$, and *skew-symmetric* if $A = -A^T$.

Determinant of a square matrix: $|A| = \sum_{j=1}^{n} a_{ij}c_{ij},\; c_{ij} = (-1)^{i+j}M_{ij}$, for the i^{th} row and M_{ij} is the ij^{th} minor; matrix A is *singular* if $|A| = 0$ and *non-singular* is $|A| \neq 0$.

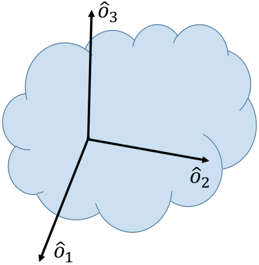
The *rank* of a matrix is the number of linearly independent columns or rows.

Trace of a matrix: sum of the diagonals, $tr(A) = \sum_{i=1}^{n} a_{ii}$.

Reference Frame: A three-dimensional vector space (e.g., Euclidian space $\mathbb{R}^3$ with a specified origin).



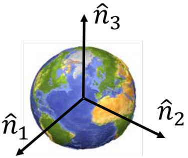
Cartesian Coordinate System (CCS): A set of mutually orthogonal basis vectors attached to the origin of a particular reference frame. The set of unit vectors is denoted $\{\hat{o}_1, \hat{o}_2, \hat{o}_3\}$.



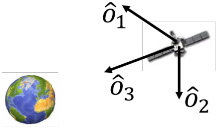
Note: We will use "right-handed" CCS in our analyses throughout this course.

Common Coordinate Systems

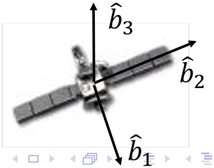
- Earth-Centered Inertial (ECI): inertially fixed; origin at center of Earth, 1 axis points to vernal equinox, 3 axis along Earth's rotation axis, and 2 axis completes the right-handed CCS.
- Orbital (or LVLH) CCS: spacecraft-fixed and rotates as spacecraft moves along orbital path; 3 axis points in nadir direction, 2 axis points in the negative orbit normal, and 1 axis completes the right-handed system.
- Body CCS: spcecraft-fixed and rotates with the spacecraft; generally the origin is at the center of mass, and axes aligned with the body principal axes.



ECI

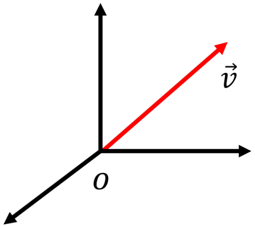


Orbital



Body

Vectors are expressed mathematically by *resolving* in a particular CCS.



- *Scalar components*: vector expressed as linear combination of basis vectors

$$\vec{v} = v_1\hat{o}_1 + v_2\hat{o}_2 + v_3\hat{o}_3$$

$$v_1 = \vec{v} \cdot \hat{o}_1, \;\; v_2 = \vec{v} \cdot \hat{o}_2, \;\; v_3 = \vec{v} \cdot \hat{o}_3$$

- *Matricial representation*: vector expressed as 3×1 matrix

$$\vec{v} = \begin{bmatrix} \hat{o}_1 & \hat{o}_2 & \hat{o}_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \Longrightarrow \boldsymbol{v}^O = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

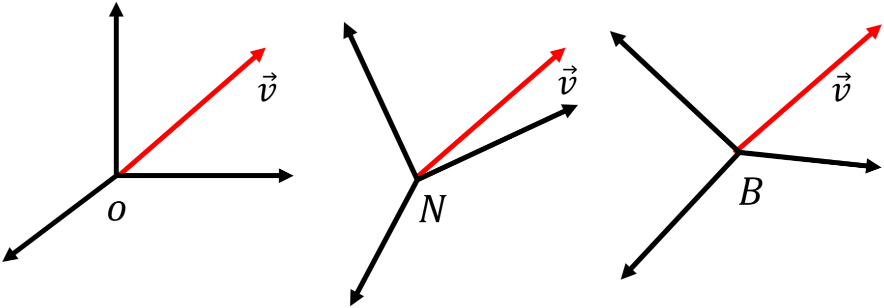
the matrix of basis vectors $[\hat{o}_1 \;\; \hat{o}_2 \;\; \hat{o}_3]$ is called a *vectrix*. Resolved vectors will be denoted as column vectors ($\boldsymbol{v}^O$, or $\boldsymbol{v}$).

<u>Note</u>: Be sure to keep track of CCS in which vector is expressed!

Representation of Vectors (cont.)

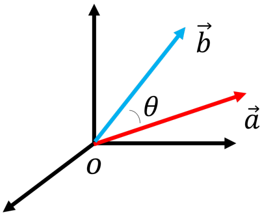
Vectors can be expressed in different CCSs but will have different components for each CCS.

$$
\begin{aligned}
\vec{v} &= v_{o1}\hat{o}_1 + v_{o2}\hat{o}_2 + v_{o3}\hat{o}_3 \\
&= v_{n1}\hat{n}_1 + v_{n2}\hat{n}_2 + v_{n3}\hat{n}_3 \\
&= v_{b1}\hat{b}_1 + v_{b2}\hat{b}_2 + v_{b3}\hat{b}_3
\end{aligned}
$$



- *Inner (or dot) product*:

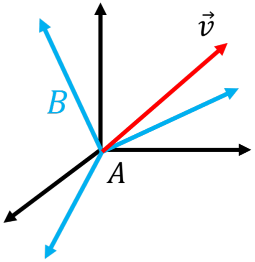
$$\begin{aligned}\vec{a}\cdot\vec{b} &= ||\vec{a}||\cdot||\vec{b}||\cos\theta \\ &= \boldsymbol{a}^T\boldsymbol{b} \\ \vec{a}\cdot\vec{b} &= \vec{b}\cdot\vec{a} \text{ (Commutative)}\end{aligned}$$



- *Vector (or cross) product*:

$$\begin{aligned}\vec{a}\times\vec{b} &= ||\vec{a}||\cdot||\vec{b}||\sin\theta\hat{k},\;\; \hat{k}\perp\vec{a} \text{ and } \hat{k}\perp\vec{b} \\ &= [\boldsymbol{a}^{\times}]\boldsymbol{b} \\ \vec{a}\times\vec{b} &= -\vec{b}\times\vec{a} \text{ (Non-commutative)}\end{aligned}$$

- Relating CCS B to A:



$$\begin{aligned}\hat{b}_1 &= C_{11}\hat{a}_1 + C_{12}\hat{a}_2 + C_{13}\hat{a}_3\\ \hat{b}_2 &= C_{21}\hat{a}_1 + C_{22}\hat{a}_2 + C_{23}\hat{a}_3\\ \hat{b}_3 &= C_{31}\hat{a}_1 + C_{32}\hat{a}_2 + C_{33}\hat{a}_3,\end{aligned}$$

where $C_{ij} = \hat{b}_i \cdot \hat{a}_j$; $\begin{bmatrix}\hat{b}_1\\ \hat{b}_2\\ \hat{b}_3\end{bmatrix} = C_{B/A}\begin{bmatrix}\hat{a}_1\\ \hat{a}_2\\ \hat{a}_3\end{bmatrix}$

- Properties of $C_{B/A}$ (or C_{BA})
  - $C_{BA}^T C_{BA} = I \implies C_{BA}^T = C_{BA}^{-1}$
  - $\det(C_{BA}) = 1$
  - C_{BA} is an *orthonormal* matrix: $C_{BA} \in SO(3)$
- Vector transformation from A to B: $\boldsymbol{v}^B = C_{BA}\boldsymbol{v}^A$

Angular Velocity of a Rotating Reference Frame

Consider two rotating reference frames with CCS A and B. The *angular velocity* of B w.r.t. A is denoted $\vec{\omega}_{B/A} = \vec{\omega}_{BA}$.

- $\vec{\omega}_{BA}$ is aligned with the instantaneous axis of rotation
- $\vec{\omega}_{BA} = -\vec{\omega}_{AB}$
- $\vec{\omega}_{CA} = \vec{\omega}_{CB} + \vec{\omega}_{BA}$

Time Derivative of a Vector

The time derivative af a vector can only be conceived w.r.t. a particular reference frame:

$$\begin{aligned}
\frac{^B d}{dt}(\vec{v}) &= {}^B\dot{\vec{v}} = \dot{v}_1\hat{b}_1 + \dot{v}_2\hat{b}_2 + \dot{v}_3\hat{b}_3 \\
\frac{^A d}{dt}(\vec{v}) &= \dot{v}_1\hat{b}_1 + \dot{v}_2\hat{b}_2 + \dot{v}_3\hat{b}_3 + v_1\frac{^A d}{dt}(\hat{b}_1) + v_2\frac{^A d}{dt}(\hat{b}_2) + v_3\frac{^A d}{dt}(\hat{b}_3)
\end{aligned}$$

where,

$$\frac{^A d}{dt}(\hat{b}_i) = \vec{\omega}_{B/A} \times \hat{b}_i$$

The *Transport Theorem*:

$$\boxed{\frac{^A d}{dt}(\vec{v}) = \frac{^B d}{dt}(\vec{v}) + \vec{\omega}_{BA} \times \vec{v}}$$

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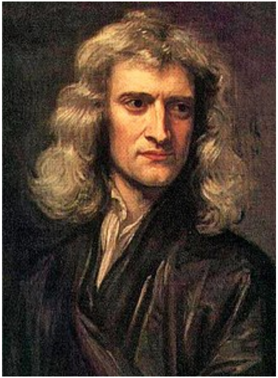
Foundations of Classical Mechanics

- Sir Isaac Newton published *Philosophiae Naturalis Principia Mathematica* in 1687, laying the foundations for classical mechanics
- Laws of motion and gravitation
- Proved Kepler's laws of planetary motion



Baseline Hypothesis:

- An inertial reference frame exists (geometric structure of Euclidian 3D space)
- An absolute time exists (i.e. at any given instant, time is equal in any point in space)

Newtons Laws of Motion

1. Law of Inertia: A particle remains in its state of rest or uniform, straight-line motion unless it is acted upon by an external force.
2. The force acting on a particle is equal to the mass of the particle times its inertial acceleration (particle mass assumed constant).

$$\vec{F} = m\vec{a} = m\ddot{\vec{r}}$$

3. For every applied force, there is an equal and opposite reaction force.

Newton's Law of Gravitation

The gravitational force between two particles of mass m_1 and m_2, separated by a distance r is given by:

$$\vec{F}_g = -\frac{Gm_1m_2}{r^3}\vec{r}$$

where G is the gravitational constant, and $\vec{r}$ is the position of m_2 relative to m_1.

Kinematics of a Particle

Kinematics is the study of motion without considering the causes (no forces involved); the focus is expressing position, velocity, and acceleration. Note: Derivatives must be taken with respect to an inertial frame to satisfy Newton's Laws, but vectors can be expressed in any CCS.

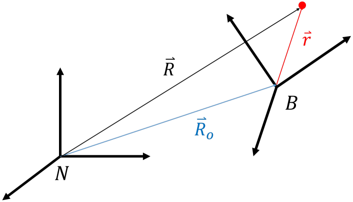
Given a particle in a moving reference frame B, with position vector $\vec{r}$, and inertial reference frame N:

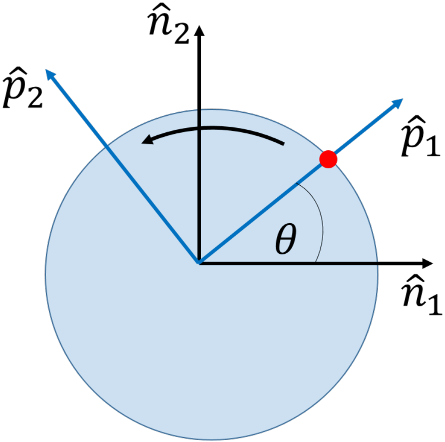
$$\vec{R} = \vec{R}_o + \vec{r},\ \vec{r} = r_1\hat{b}_1 + r_2\hat{b}_2 + r_3\hat{b}_3$$

$$\frac{{}^N d}{dt}(\vec{R}) = \dot{\vec{R}} = \dot{\vec{R}}_o + \frac{{}^B d}{dt}(\vec{r}) + \vec{\omega}_{BN} \times \vec{r}$$

$$\frac{{}^N d^2}{dt^2}(\vec{R}) = \ddot{\vec{R}} = \ddot{\vec{R}}_o + \frac{{}^B d^2}{dt^2}(\vec{r}) + \dot{\omega}_{BN} \times \vec{r}$$

$$+2\left(\vec{\omega}_{BN} \times \frac{{}^B d}{dt}(\vec{r})\right) + \vec{\omega}_{BN} \times \vec{\omega}_{BN} \times \vec{r}$$





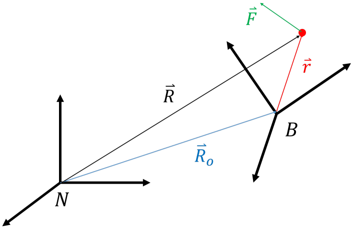
Write the position, velocity, and acceleration of the particle, expressed in the P CCS.

Dynamics of Particle

Dynamics is the study of motion with consideration of its causes (with forces and torques); application of Newton's Second Law.

Let $\vec{F}$ be the resultant of all forces acting on a particle; the equation of motion is written:

$$m\ddot{\vec{R}} = \vec{F}$$



D'Alembert's Principle: alternate form of Newton's Second Law:

$$\vec{F} - m\ddot{\vec{R}} = 0$$

Momentum, Torque, and Energy

Linear Momentum of a particle: $\vec{p} = m\dot{\vec{r}} = m\vec{v}$.

Angular Momentum of a particle about an arbitrary point O:
$\vec{H}_O = \vec{r} \times m\vec{v} = \vec{r} \times m\vec{p}$, where $\vec{r}$ is the particle's position relative to point. O

Torque (or *moment*) about point O: $\vec{M}_O = \vec{r} \times \vec{F}$.

Kinetic Energy: $T = {}^1\!/{}_2 m\dot{\vec{r}} \cdot \dot{\vec{r}} = {}^1\!/{}_2 m\vec{v} \cdot \vec{v}$.

A given force is said to be *conservative* if it depends only on the position vector ($\vec{r}$), and can be written as the gradient of the *potential energy*, V: $\vec{F} = -\nabla V$.

Conservation of Energy: If a particle is acted upon only by conservative forces, the total energy (E) remains constant ($E = T + V = constant \implies \dot{E} = 0$).